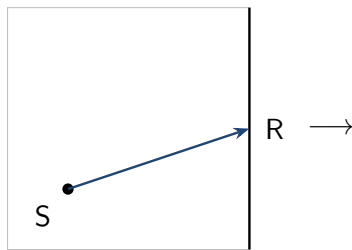
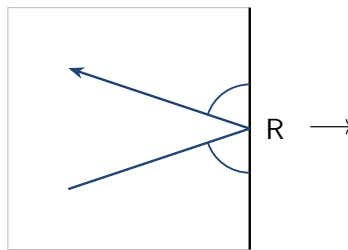


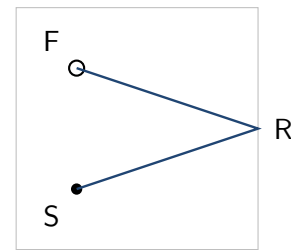
Each path travels straight between walls. At a bounce, the two angles to the wall are equal. A path stops if it hits a corner. The walls are left L, right R, bottom B and top T.



Incoming



Equal angles

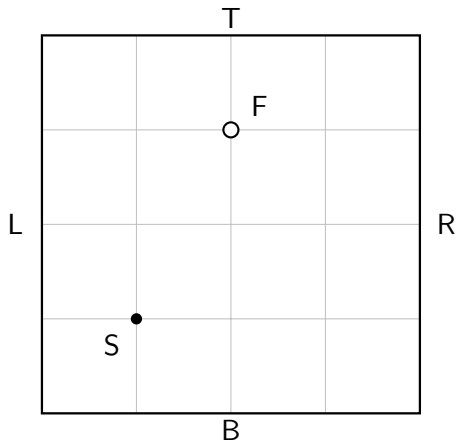


Word: R

Write wall names in the order touched: LR means left, then right.

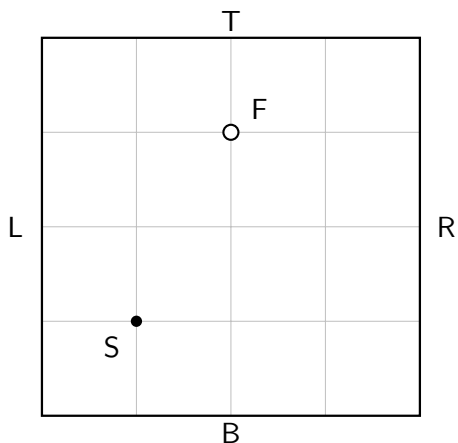
Problem 1: Find every wall word that gets from S to F with exactly two bounces. Find the words for both finishes. A word may use the same wall twice, if a path can do it. Draw each path with a ruler.

Finish A



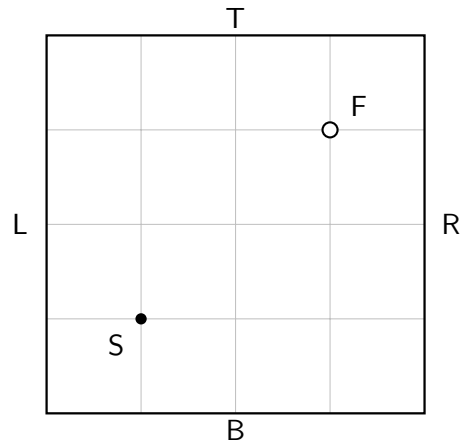
Word: _____

Finish A



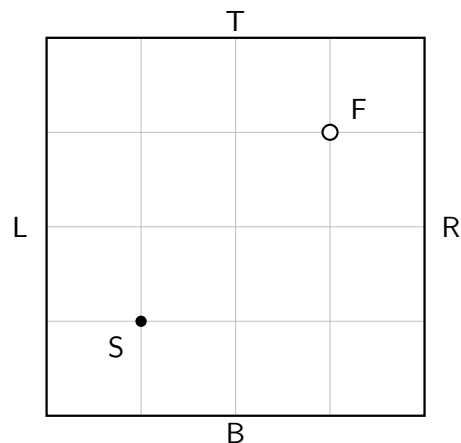
Word: _____

Finish B



Word: _____

Finish B



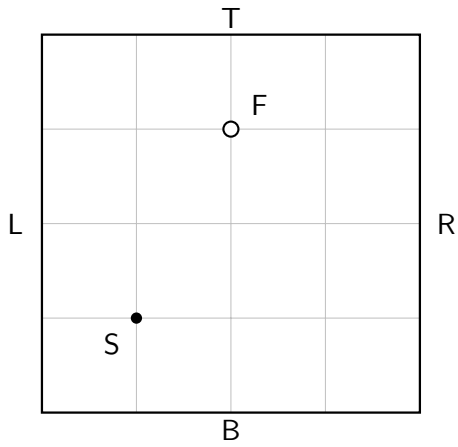
Word: _____

Extra workspace for Problem 1.

Finish A words: _____

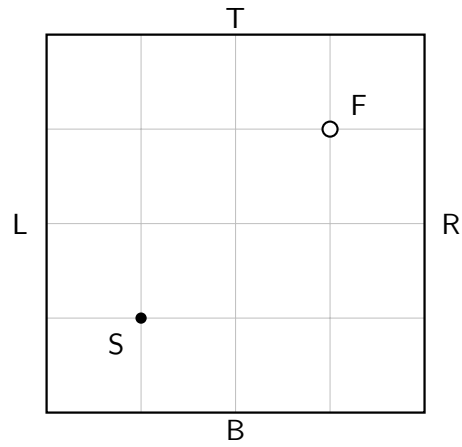
Finish B words: _____

Finish A



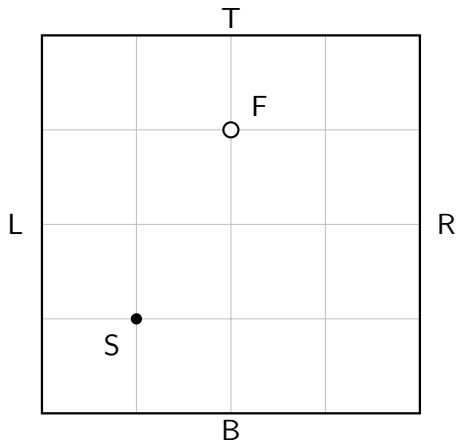
Word: _____

Finish B



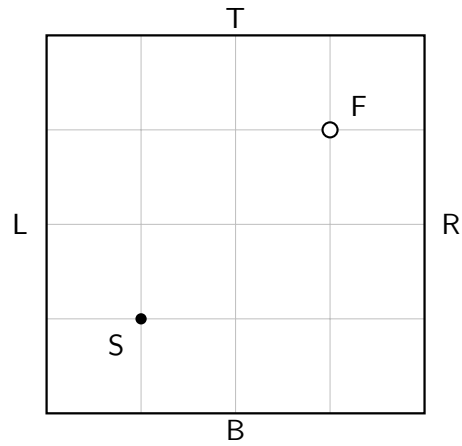
Word: _____

Finish A



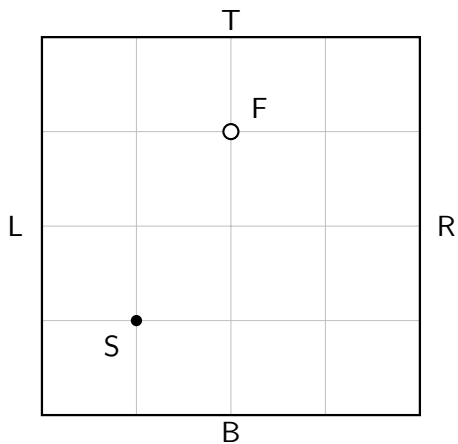
Word: _____

Finish B



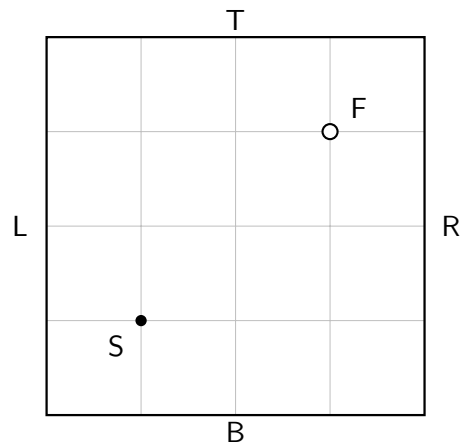
Word: _____

Finish A



Word: _____

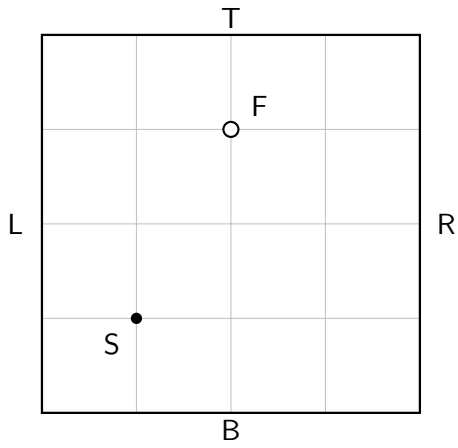
Finish B



Word: _____

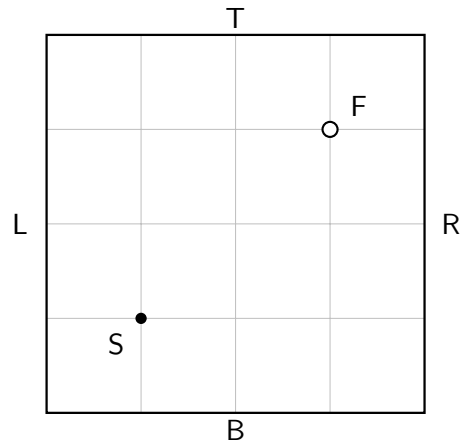
Extra workspace for Problem 1.

Finish A



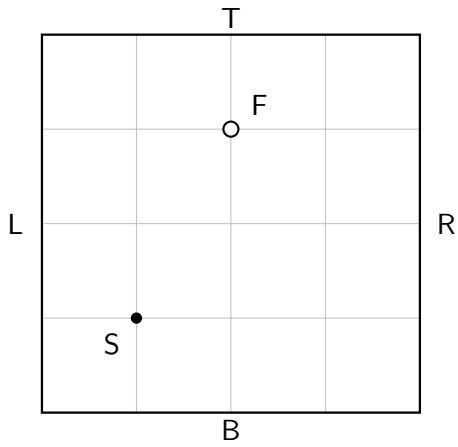
Word: _____

Finish B



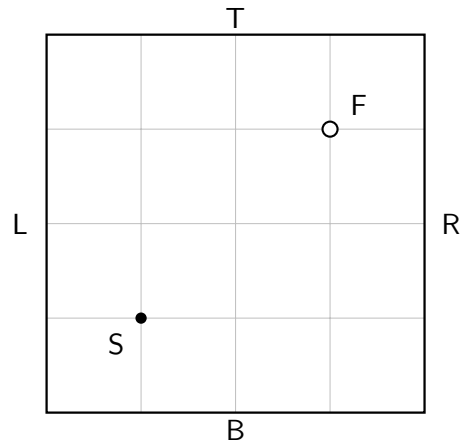
Word: _____

Finish A



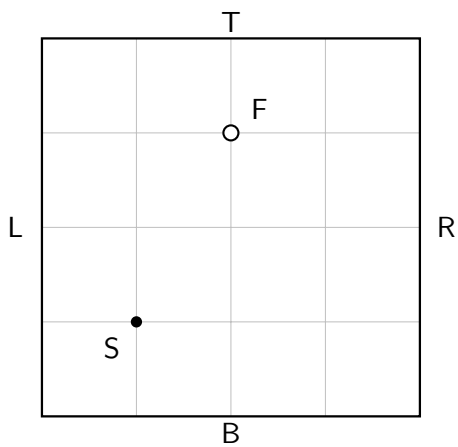
Word: _____

Finish B



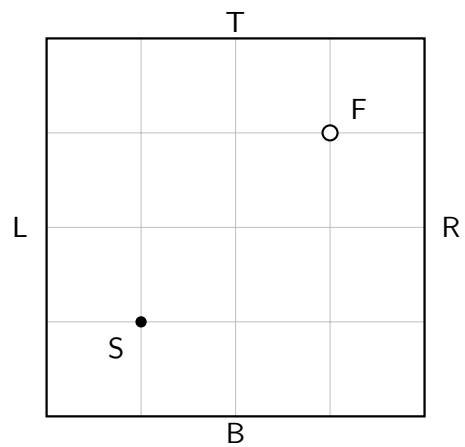
Word: _____

Finish A



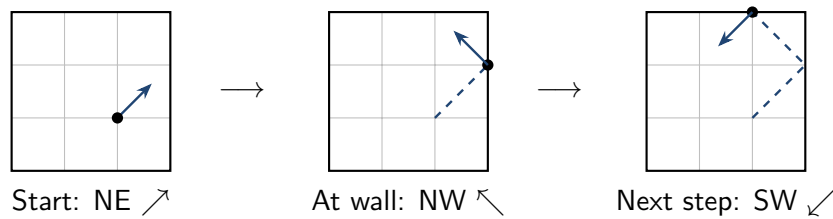
Word: _____

Finish B

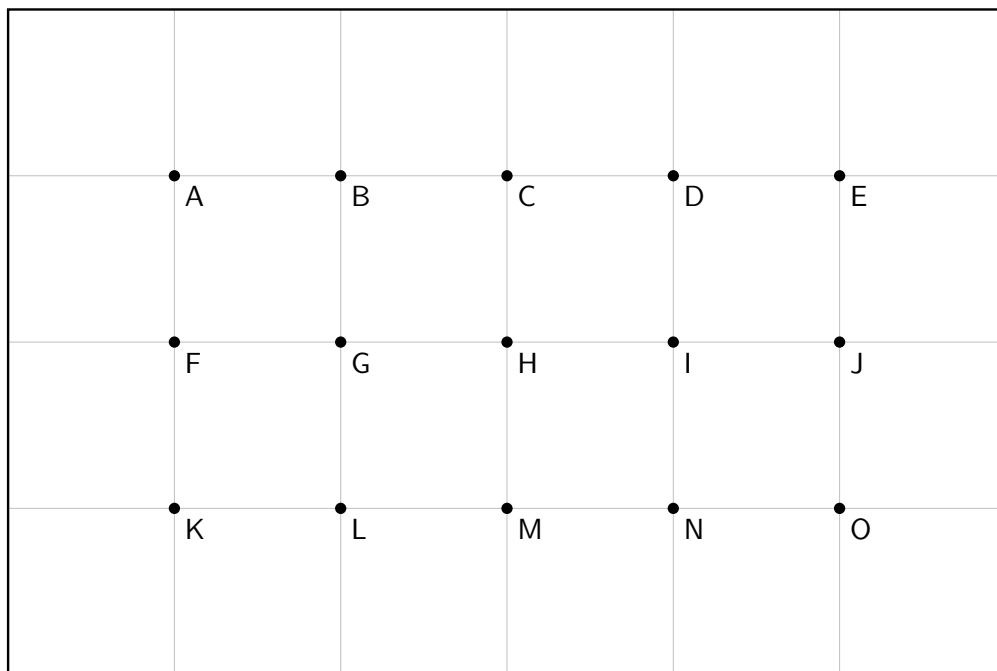


Word: _____

For diagonal paths, move one square across and one square up or down each step. Keep the counter's arrow pointing along its next move. At a wall, turn the arrow away from that wall. Stop at a corner.



Problem 2: Start at each lettered dot, with the arrow pointing NE ↗. Which starts reach a corner? Which return to their starting dot with the arrow pointing NE ↗? Mark C for a corner and L for a loop. For each loop, find the number of diagonal steps until that first return.



A _____	B _____	C _____	D _____	E _____
F _____	G _____	H _____	I _____	J _____
K _____	L _____	M _____	N _____	O _____

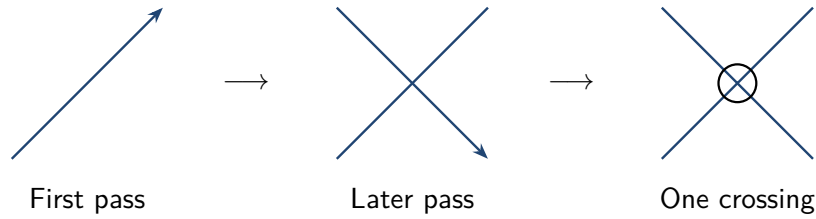
Loop starts and their step counts: _____

Extra workspace for Problem 2.

Starting letter: _____ Loop steps: _____

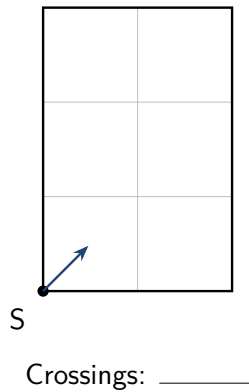
Starting letter: _____ Loop steps: _____

For diagonal paths, move one square across and one square up or down each step. A crossing is one place inside the board where two pieces of the path form an X. Wall hits and corners do not count.

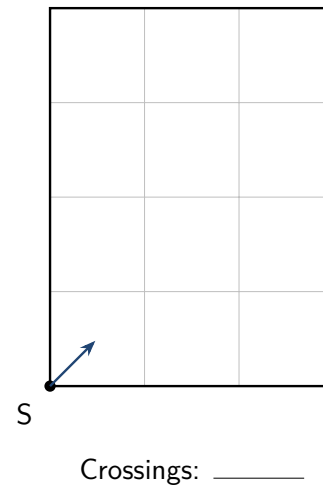


Problem 3: Start at S heading NE ↗ and draw each diagonal path until it reaches a corner. Circle its crossing points. Which boards have the same number of crossings? Find a rectangle with exactly ten crossings on the next page.

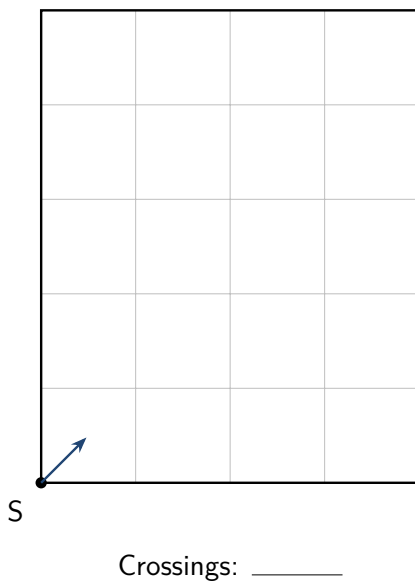
2 wide, 3 high



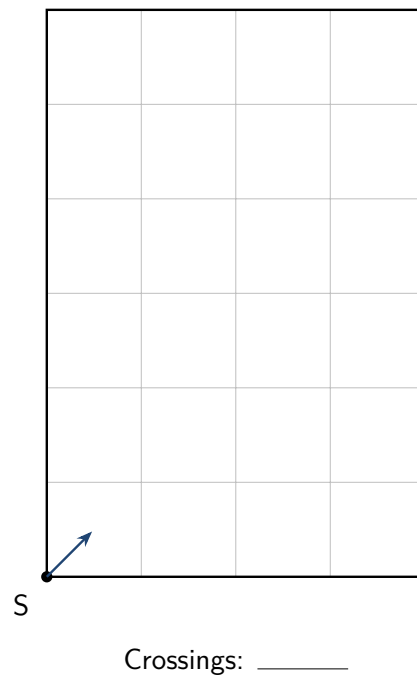
3 wide, 4 high



4 wide, 5 high



4 wide, 6 high



Extra workspace for Problem 3. Outline your rectangle on the grid.



My rectangle: _____ wide, _____ high.

Crossings: _____